ACADEMIC RESEARCH PROJECT (MBA-WILP)

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE PILANI

EC1: 3 Phase Mini Project Plan

Applied Data Science (S1-25_MBACCZG581)

PHASE 3 End-Term Phase – Step 10 & 11

Exploratory Analysis of Factors Influencing Housing Prices — Boston Housing Dataset

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Executive Summary

Context

Housing prices are influenced by multiple factors such as crime rates, number of rooms, accessibility to highways, and school quality. Understanding these relationships is critical for real estate companies, policymakers, and urban planners. The Boston Housing dataset provides a rich set of variables that reflect socio-economic and environmental aspects of neighborhoods. This project uses this dataset as a case study to demonstrate how data-driven insights can support better decision-making.

Acknowledgement

We sincerely thank BITS Pilani Faculty (Shree Gokhale Jayanti) for providing this opportunity to apply data science concepts to real-world problems. We also acknowledge the support of our peers and the availability of public datasets from UCI Machine Learning Repository and Kaggle, which made this analysis possible.

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Abbreviation Table

Abbreviation	Full Form
CRIM	Crime rate per capita
ZN	Residential land zoned for large lots (%)
INDUS	Non-retail business acres (%)
CHAS	Charles River dummy variable
NOX	Nitric oxide concentration
RM	Average number of rooms per dwelling
AGE	% of houses built before 1940
DIS	Distance to employment centers
RAD	Index of highway accessibility
TAX	Property tax rate
PTRATIO	Pupil-teacher ratio
B	Race-based calculation
LSTAT	% of lower-status population
MEDV	Median house value ($1000s)

1. Business report / Executive summary

1.1 Summary Explanation

- This project studies what drives residential property prices in Boston using a public dataset.
- We followed BITS EC1's three-phase approach: (1) define the problem and dataset, (2) clean and explore the data to find patterns, and (3) finalize insights, ethics, and recommendations for decision-makers.
- Across the analysis, four signals stand out.
  - More rooms (RM) is linked with higher price (MEDV);
  - Higher crime (CRIM) is linked with lower price;
  - Better school attention (lower PTRATIO) aligns with higher price;
  - Socio-economic vulnerability (LSTAT) is negatively related to price.
- These are practical and explainable drivers that real-estate teams and policy makers can use for decisions. (Patterns summarized from Phase 2 EDA.)
- The insights help three groups.
  - Developers/agents can prioritize features that raise value (room count, school advantage, safer locations).
  - Policy makers can see how safety and schooling investments support neighbourhoods and long-term market health.
  - Buyers/investors can use these signals as a simple due-diligence checklist. (Scope and roles are consistent with EC1's non-ML exploratory focus.)

1.2 Problem & Dataset (Phase 1 Recap)

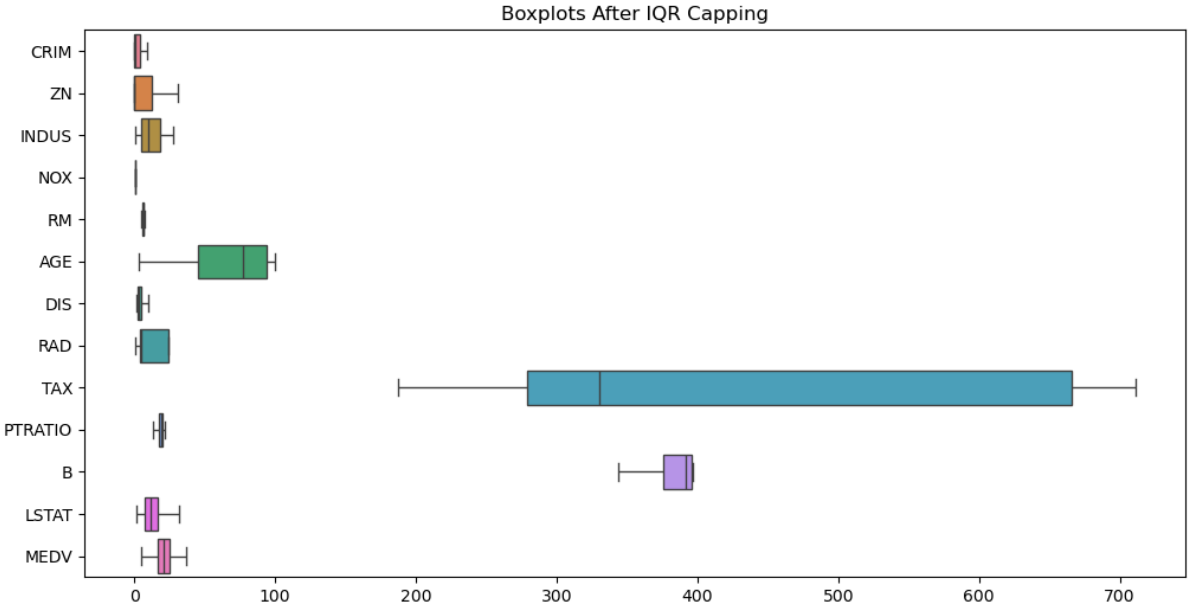
- Problem. Understand which neighborhood and property factors go together with higher or lower housing prices; no predictive modeling—only exploration and interpretation.
- Dataset. Boston Housing (UCI ML via Kaggle). 506 rows × 14 columns; mostly numeric, CHAS is categorical; no missing values; price (MEDV) is capped at 50.

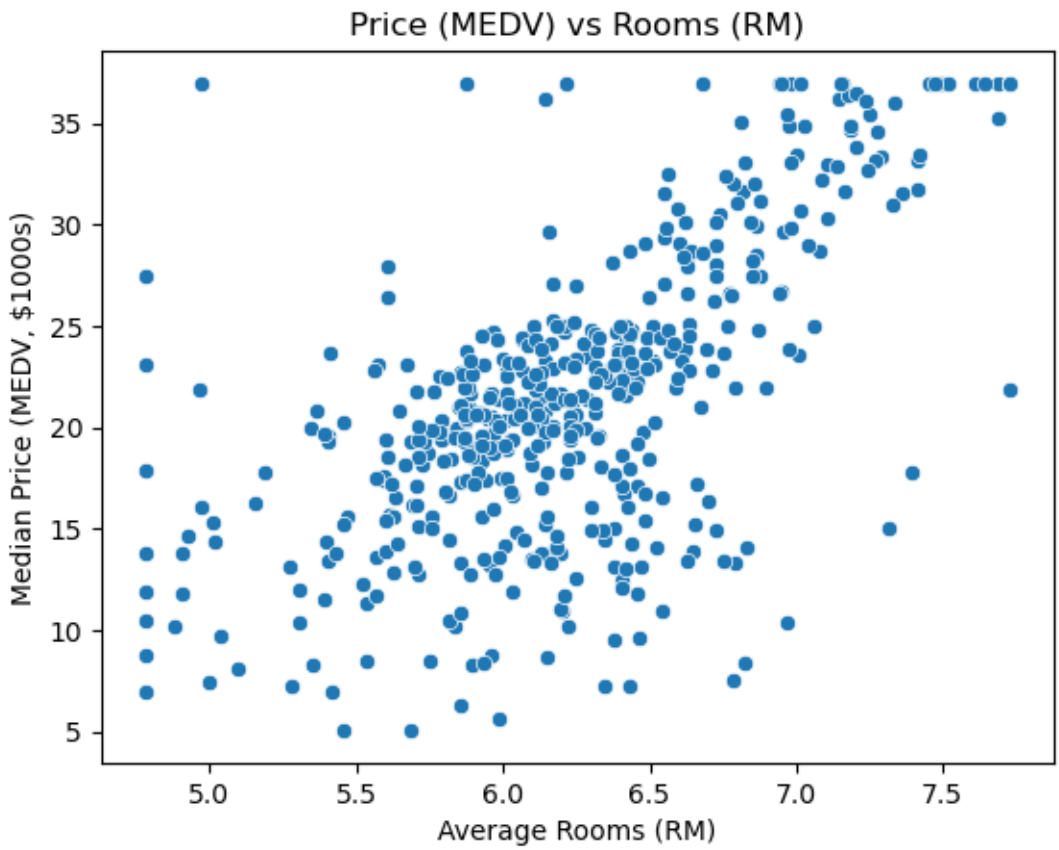
1.3 Method (Phase 2 Recap)

- Cleaning. Added headers, set CHAS categorical, checked missing values (none), treated outliers using IQR capping to stabilize visuals without dropping records.
- Exploration. Descriptive stats, boxplots, scatter plots (RM/CRIM/PTRATIO vs MEDV), distribution of MEDV, and correlation heatmap; each interpreted with a brief business meaning.

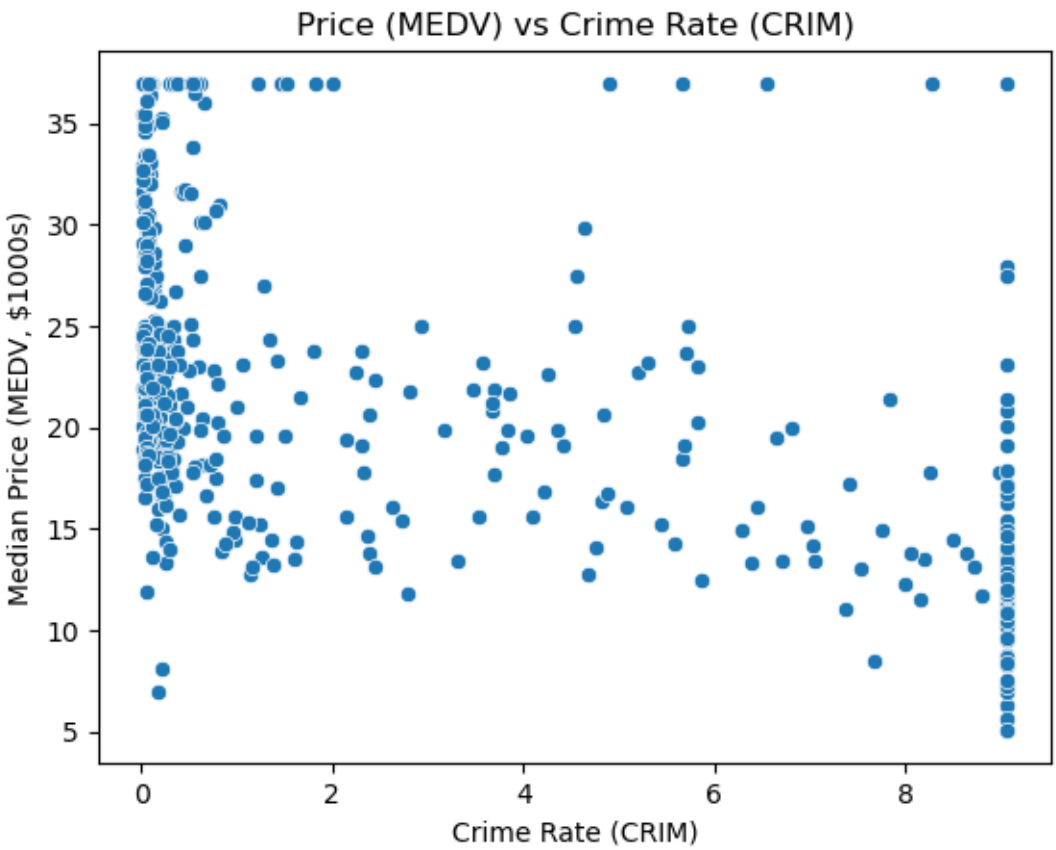
2. Key Findings (What the visuals say)

- Rooms (RM) vs Price (MEDV): clear upward pattern — larger homes command higher prices.
  Implication: Emphasize room count and layout in listings and valuation talking points.
- Crime (CRIM) vs Price: downward relationship — safer areas tend to be valued higher.
  Implication: Neighborhood safety is a core pricing KPI and marketing lever.
- School Attention (PTRATIO) vs Price: lower PTRATIO aligns with higher price.
  Implication: School metrics belong on property brochures targeting families.
- Socio-economic (LSTAT) vs Price: strong negative link with MEDV.
  Implication: Community uplift and local amenities matter for long-term value.
- Price Distribution: capped at 50 (top-coding).
  Implication: Interpret high-end segment cautiously; true luxury prices may be understated.
- Correlation Heatmap: RM (+), LSTAT (–), CRIM (–), PTRATIO (–) are the most influential associations.
- Implication: These serve as the short-list of drivers for communication and tracking.
- MEDV is top-coded at 50.0 in the original dataset. Additional IQR winsorization further compresses the upper tail. A quick "before vs after" check in the notebook confirmed that the direction of key relationships (RM, CRIM, PTRATIO, LSTAT) remains unchanged.

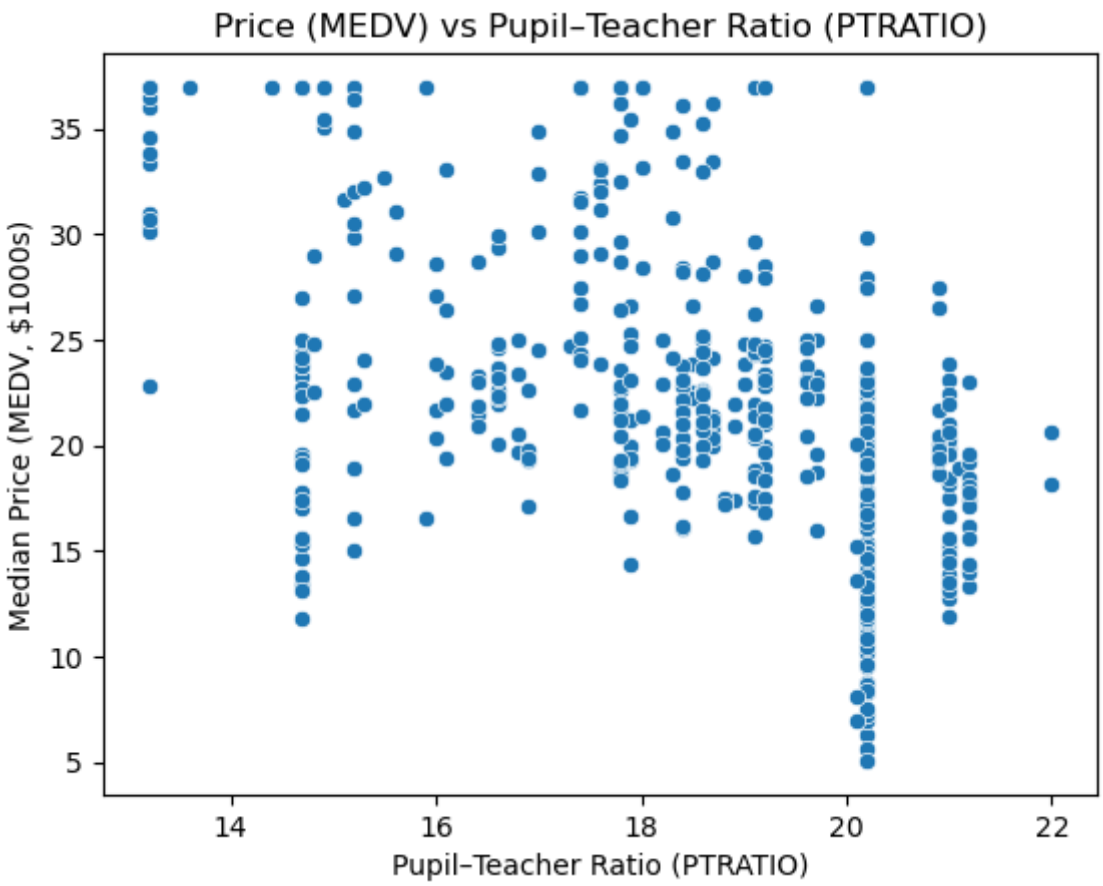




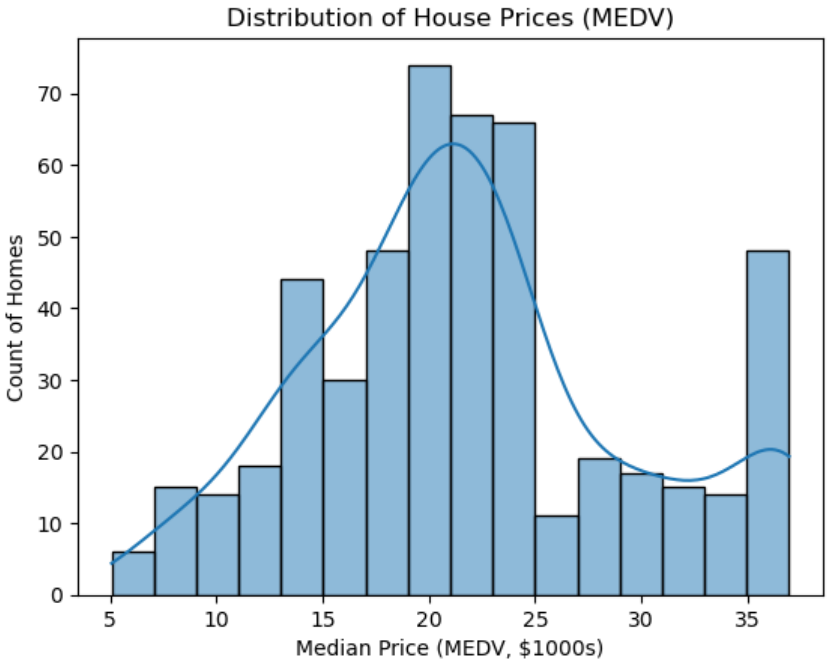
- **RM vs MEDV (rooms vs price).**
- Positive association: higher RM aligns with higher MEDV.
- Business note: Larger, well-laid-out homes tend to command a premium in this dataset.

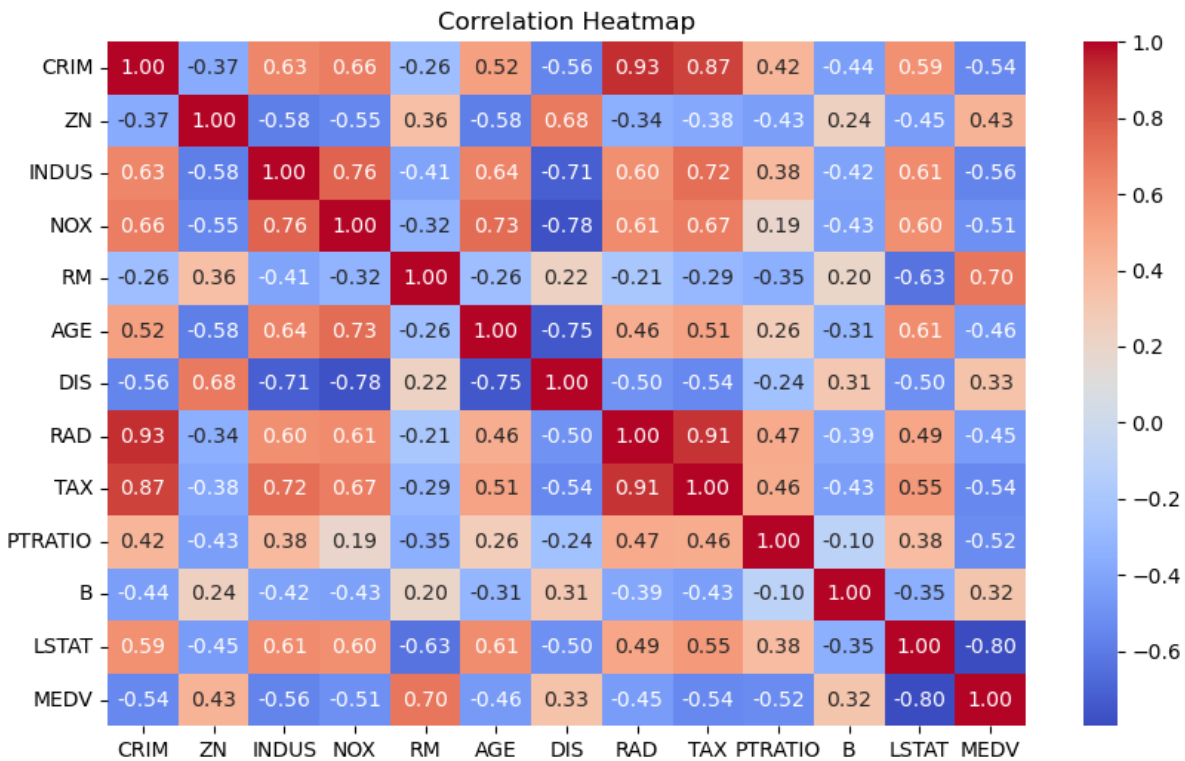


- **CRIM vs MEDV (crime vs price).**
- Negative association: higher CRIM neighborhoods show lower MEDV.
- Business note: Neighborhood safety is a core pricing lever



- **PTRATIO vs MEDV (school attention vs price)**
- Lower PTRATIO tends to align with higher MEDV.
- Business note: school capacity/attention appears valued by buyers





- **Heatmap (correlation, winsorized)**
- Top MEDV associations: RM (+), LSTAT (–), CRIM (–), PTRATIO (–).
- These are the simplest, consistent signals to carry into the business summary

3. Recommendations

For developers/agents:

- Highlight RM (room count) and functional layouts in pricing conversations.
- Promote school proximity / low PTRATIO areas as “education advantage” zones.
- Stress neighbourhood safety where metrics are strong; offer risk disclosure where needed.

For policy makers:

- Safety programs and community policing can indirectly lift property values.
- Investments in public education (teacher staffing, facilities) support housing markets.

For buyers/investors:

- Use a simple checklist: RM↑, PTRATIO↓, CRIM↓, LSTAT↓ → stronger pricing zones; the opposite → higher risk.

4. Ethical Considerations

- Sensitive attributes. LSTAT reflects socio-economic vulnerability; do not use it to stigmatize communities or enable discriminatory pricing.
- Transparency. Document cleaning steps (IQR capping, types) and communicate the capped MEDV limitation clearly.
- Responsible use. Apply insights for community uplift (safety, education), not exclusion.
- Academic integrity. Cite public sources; keep writing original; declare AI assistance honestly.
- Sensitive fields: LSTAT (socio-economic correlate) and B (race-based calculation) are discussed only as correlates in a historic academic dataset. They are not used to target, rank, or exclude communities.
- Transparency: This report clearly documents data preparation (headers, types, winsorization), acknowledges MEDV top-coding, and states the contextual scope (historic Boston dataset).

5. Limitations & Future Work

- Historic, Boston-specific dataset: findings are contextual, not universal. .
- Outcome censoring: MEDV is top-coded at 50.0; the high-end segment is understated.
- Non-predictive by design: No ML modelling in EC1.
- Future work could test robust regressions or add geo/context features (if permitted).

6. Acknowledgments

- We acknowledge the UCI ML Repository / Kaggle dataset and BITS Pilani EC1 guidance for the three-phase structure and deliverables.
- 8. IP & AI Tools Usage (Paste as a separate small box)
- IP / Copyright Notice. This report is prepared solely for academic purposes. The dataset is public; any proprietary references have been generalized.
- AI Tools Usage Declaration. AI tools were used to assist in organizing ideas and language. The final analysis, choices, and edits are by the authors.

7. Conclusion

- This project, completed across three phases, identified four clear factors linked with Boston housing prices: RM (higher rooms → higher price), CRIM (higher crime → lower price), PTRATIO (lower ratio → higher price), and LSTAT (higher vulnerability → lower price). The cleaned dataset, structured EDA, and refined visuals consistently supported these relationships.
- These insights help **developers** highlight value-adding features, **policymakers** focus on safety and education improvements, and **buyers** evaluate neighbourhoods more effectively. Dataset limits (historic data, MEDV capped at 50) were acknowledged, and sensitive fields were interpreted responsibly.
- Overall, the analysis provides a simple, practical understanding of what drives housing value in this dataset—without predictive modelling.

_____Report Ends Here_____

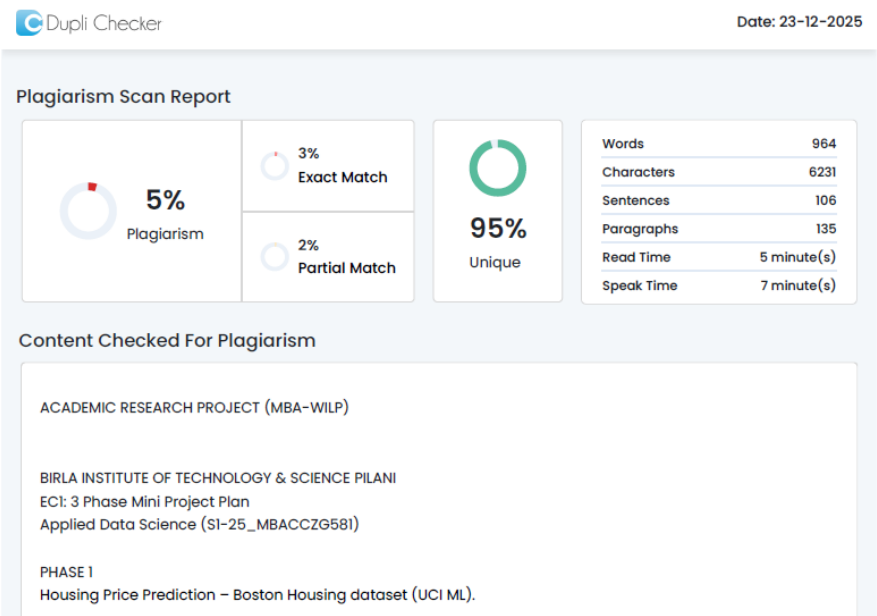
8. Academic Declarations

References

- References • Harrison, D., & Rubinfeld, D. L. (1978). Boston Housing Dataset. UCI ML Repository: https://archive.ics.uci.edu/dataset/2/boston+housing .
- Kaggle. (n.d.). Boston Housing Dataset. https://www.kaggle.com/datasets Note:
- The raw Kaggle file is space-separated and does not include headers; headers were added based on UCI documentation.

Plagiarism Declaration

"I hereby declare that this assignment is my original work and has not been copied from any other source. All references and supporting materials have been appropriately acknowledged."



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